

WORLD AIR QUALITY REPORT 2025

Summary, Key Findings, Original Material & Summarization Method

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Purpose of this document: The supplied assignment material about IQAir's World Air Quality Report 2025 has been condensed into a short, structured summary. The original material is retained later in this document so that the reader can compare the source content with the summarized version.

1. Executive Summary

The World Air Quality Report 2025, as described in the supplied material, assesses global PM2.5 pollution across 9,446 cities in 143 countries. The report emphasizes that fine particulate matter remains a major public-health and economic risk. Only 14% of global cities met the WHO annual PM2.5 guideline of $5 \mu\text{g}/\text{m}^3$, while 130 of 143 countries and territories did not meet the guideline. Asia remained the most affected region. India showed a modest improvement in population-weighted PM2.5, but continued to rank among the world's most polluted countries, with Delhi and several Indian cities recording very high concentrations.

2. Summary — Key Points

- Global coverage: The report assesses PM2.5 air pollution across 9,446 cities in 143 countries using data from more than 40,000 monitoring stations and sensors.
- WHO benchmark: PM2.5 is measured in $\mu\text{g}/\text{m}^3$ and compared with the WHO annual guideline of $5 \mu\text{g}/\text{m}^3$.
- Global compliance: Only 14% of global cities met the WHO guideline in 2025, compared with 17% in 2024.
- Countries affected: 130 of 143 countries and territories failed to meet the stated safe guideline.
- Regional pattern: Asia remained the most affected region, and the supplied material states that all top 25 most polluted cities were in India, Pakistan, or China.
- Most polluted country: Pakistan ranked first, followed by Bangladesh and Tajikistan, according to the supplied material.
- Major contributors: Wildfires and industrial emissions were identified as important contributors to PM2.5 pollution.
- Overall message: Air pollution remains a significant health and economic risk requiring continuous monitoring and coordinated mitigation.

3. India — Key Points

- National PM2.5: India's population-weighted PM2.5 concentration was reported as $48.9 \mu\text{g}/\text{m}^3$ in 2025, a 3% decrease from 2024 and 10% from 2023.
- Global ranking: Despite improvement, India ranked 6th globally in air pollution in the supplied material.
- New Delhi: Delhi recorded $82.2 \mu\text{g}/\text{m}^3$, described as a three-year low, while remaining the world's most polluted capital.
- Most polluted Indian city: Loni in Uttar Pradesh was reported at $112.5 \mu\text{g}/\text{m}^3$, followed by Bijnor and Delhi.
- Indian cities in global ranking: 66 of the 100 most polluted cities globally were reported to be in India.
- NCAP focus: The supplied material states that 64% of National Clean Air Program funding is directed towards road-dust control, with comparatively less focus on biomass burning, vehicles, and industrial pollution.
- Monitoring: India was described as having an extensive monitoring network covering 259 cities, giving broader visibility into pollution levels.

4. Short Paragraph Summary

The supplied World Air Quality Report 2025 material shows that PM2.5 pollution remains a serious global problem. Most countries and many cities exceeded the WHO annual guideline, with Asia experiencing particularly high pollution. India recorded a moderate improvement in average PM2.5 levels, but pollution remained severe, especially in Delhi and several northern and industrial regions. The findings reinforce the importance of monitoring, controlling emissions, addressing road dust and other sources, and maintaining coordinated long-term air-quality policies.

5. How the Summary Was Created

The summary was created by reducing the supplied material while preserving the facts needed to understand the report's purpose, scale, major findings, India-specific findings, and overall conclusion. The process focused on information value rather than simply shortening every sentence.

Step	What was considered
1. Identify the topic	The central topic was identified as global PM2.5 air pollution and its health/economic significance.
2. Identify scope	The geographic and data scope was retained: 9,446 cities, 143 countries, and more than 40,000 monitoring stations/sensors.
3. Extract key numbers	Important statistics were retained, including the 14% compliance rate, 130/143 countries, India's 48.9 µg/m ³ , Delhi's 82.2 µg/m ³ , and Loni's 112.5 µg/m ³ .
4. Separate global and India findings	Global findings and India findings were grouped separately so the reader can understand both levels of the report.
5. Preserve comparisons	Changes from 2024/2023 and comparisons with the WHO guideline were retained because they show direction and severity.
6. Remove repetition	Repeated statements about pollution being dangerous, urgent, or requiring action were combined rather than repeated.
7. Simplify wording	Long explanatory sentences were converted into shorter statements and bullet points without changing their intended meaning.
8. Retain causes and response	Wildfires, industrial emissions, monitoring, NCAP funding, and mitigation were retained because they explain causes and possible responses.
9. Create a conclusion	The final short paragraph combines the main global and Indian findings into one overall message.

6. What Was Considered Important for the Summary?

- Purpose: What the report measures and why PM2.5 matters.
- Scale: How many cities/countries and monitoring sources were covered.
- Benchmark: The WHO annual PM2.5 guideline used for comparison.
- Major global findings: Compliance rate, countries exceeding the guideline, regional concentration, and major sources.
- India findings: National trend, international ranking, Delhi, most polluted Indian city, and number of Indian cities in the global top 100.
- Policy/monitoring: NCAP funding focus and the monitoring network.
- Overall implication: The continuing need for monitoring, mitigation, and coordinated action.

7. Original Report Material Provided in the Assignment

About the World Air Quality Report 2025

The World Air Quality Report 2025, published by IQAir, presents a comprehensive assessment of air pollution across the globe, with a focus on PM2.5, fine particulate matter that poses significant risks to human health.

PM2.5 particles are smaller than 2.5 micrometers, penetrate deep into the lungs and bloodstream, and are linked to cardiovascular, respiratory, and developmental diseases, particularly in children.

The report analyses data from 9,446 cities across 143 countries, sourced from over 40,000 regulatory monitoring stations and low-cost sensors, contributed by governments, research institutions, non-profit organizations, private entities, and citizen scientists.

PM2.5 concentrations are reported in micrograms per cubic meter (µg/m³), benchmarked against World Health Organization (WHO) annual guidelines, facilitating international comparison and risk assessment.

Compared to prior editions, the 2025 report expanded geographic coverage, incorporating previously under-monitored regions and cities, thereby enhancing the global understanding of air quality patterns.

Global Findings of World Air Quality Report 2025

- Only 14% of global cities met the WHO annual PM2.5 guideline of 5 µg/m³ in 2025, down from 17% in 2024.
- 130 out of 143 countries and territories failed to meet safe air quality standards.
- Asia remains the worst-affected region, with all of the world's top 25 most polluted cities located in India, Pakistan, or China.
- Pakistan emerged as the most polluted country, followed by Bangladesh and Tajikistan.
- Wildfires, driven by climate change, and industrial emissions contributed significantly to global PM2.5 pollution, releasing about 1,380 megatons of carbon from biomass burning.
- Only 13 countries/territories globally met the WHO guideline, mostly in Latin America, Caribbean, and small islands.
- The report highlights air pollution as a major public health and economic risk, requiring continuous monitoring and coordinated action.

World Air Quality Report 2025 Findings Related to India

- India's population-weighted PM_{2.5} concentration fell to 48.9 µg/m³ in 2025, a 3% decrease from 2024 and a 10% decrease from 2023.
- Despite improvements, India ranks 6th globally in air pollution.
- New Delhi recorded 82.2 µg/m³, a three-year low, yet remains the most polluted capital in the world.
- The most polluted city in India was Loni (Ghaziabad district, UP) with 112.5 µg/m³, followed by Bijnor (Uttar Pradesh) and Delhi.
- 66 of the 100 most polluted cities globally are in India, reflecting the severity of the problem.
- India's National Clean Air Program (NCAP) directs 64% of funding to control road dust, with less focus on biomass burning, vehicular emissions, and industrial pollution.
- India's extensive air quality monitoring network (259 cities) provides more comprehensive data than neighboring countries, but also exposes the widespread scale of pollution.

8. Important Note

The original material in this document is reproduced from the assignment text supplied by the user. The summary is based on that supplied material and is intended to demonstrate summarization, information selection, and concise report writing. It is not an independent fact-check of every claim in the source material.